
Residual Migration Phase 0: A Headroom-Limited Residual-Clipping Gate

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Abstract

This draft packet reports the first Residual Migration Phase 0 gate after the checker returned `PASS_RM_PHASE0_RETHINKING_REPLICATES` for `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/checker_result.json`. The measured accuracy drop after residual 95th-percentile clipping is 0.0 with bootstrap 95% CI [0.0, 0.0], clearing the preregistered replicates threshold that the CI upper bound be below 0.015 according to `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/checker_result.json` and `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/bootstrap_ci.json`. This is a mechanical Phase 0 PASS, not a paper-ready positive method: baseline accuracy is 0.0 and ablation accuracy is 0.0 in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json`, so the zero drop cannot support a capability-preservation claim. Phase 1 must add headroom and oracle safeguards before any residual-quantization recipe claim is made.

1 Introduction

Residual Migration asks whether high-magnitude residual values can be clipped or migrated without destroying downstream reasoning accuracy. The Phase 0 surface is intentionally narrow: it tests whether a severe residual clipping ablation causes a measurable exact-match accuracy drop on 12 frozen AIME-2025 traces from `ibm-granite/granite-4.0-h-tiny`. The run directory is `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z`, with artifact completeness recorded in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/artifact_check.json`.

The current paper status is not COLM- or ICLR-ready. The checker PASS says that this frozen packet did not show an accuracy drop under the preregistered replicates rule. It does not show preserved capability, because the unablated baseline answered zero of the 12 prompts correctly. That headroom failure is the central limitation of this draft and is recorded in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json` and `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/checker_result.json`.

2 Phase 0 Method

The Phase 0 model is `ibm-granite/granite-4.0-h-tiny` at HuggingFace snapshot commit `791e0d3d28c86e106c9b6e0b4cecdde0375b6124`, recorded in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/model_provenance.json` and repeated in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json`. The prompt packet contains deterministic AIME-2025 indices 0–11 with prompt SHA `sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd`, recorded in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/prompt_manifest.json` and `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json`.

Quantity	Phase 0 value
Checker decision	PASS_RM_PHASE0_RETHINKING_REPLICATES
Artifact complete	true
Baseline accuracy	0.0
Ablation accuracy	0.0
Accuracy drop	0.0
Bootstrap 95% CI	[0.0, 0.0]
Preregistered pass-replicates threshold	CI upper < 0.015
Model	ibm-granite/granite-4.0-h-tiny
Model snapshot commit	791e0d3d28c86e106c9b6e0b4cecdde0375b6124
Prompt SHA	sha256: 2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd
Generation artifact SHA	sha256: 5ce5d78cd10ec8a5fe92502d23624919fa286900f3ca769fd4f6979457eee8ad
Trace count	12
Generation length	frozen 2048 max_new_tokens

Table 1: Residual Migration Phase 0 gate result. Decision, artifact completeness, threshold, and reasons are from experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/checker_result.json and experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/artifact_check.json. Accuracy, bootstrap, model, prompt, and generation SHA values are from experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json, experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/bootstrap_ci.json, experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/model_provenance.json, experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/prompt_manifest.json, and experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/artifact_hashes.json.

Generation length is frozen at 2048 max_new_tokens; the generation artifact is experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/generations.jsonl with SHA sha256:5ce5d78cd10ec8a5fe92502d23624919fa286900f3ca769fd4f6979457eee8ad in experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json and experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/artifact_hashes.json.

The ablation attaches a forward pre-hook at every discovered transformer layer. For each layer and token position, residual values whose absolute magnitude exceeds that layer-position 95th percentile are clipped to the threshold while preserving sign. The ablation definition and per-layer thresholds are stored in experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/ablation_config.json.

3 Results

The checker reason is exactly “replicates pass: ci_high=0.00000000 < 0.015” in experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/checker_result.json. The aggregation is prompt-level binary exact-match correctness and a gate-level mean baseline correctness minus mean ablation correctness across 12 prompts, as recorded in experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json.

4 Interpretation

The saturated Phase 0 fact is narrow: under this exact frozen packet, the 95th-percentile residual clipping ablation produced no measured accuracy drop. The live scientific branch is not yet a positive method, because the baseline also had no correct answers. A zero drop from 0.0 baseline accuracy cannot distinguish harmless residual migration from an evaluation surface with no headroom.

The highest-priority next branch is Phase 1 with headroom and oracle safeguards. Before any residual-quantization recipe claim, the evaluation must include a surface where the baseline succeeds often enough for a drop to be meaningful, an oracle or diagnostic showing recoverable task headroom, paired uncertainty, and a strict separation between same-family replication and cross-family falsification.

5 Limitations

This packet should not be cited as evidence that residual clipping preserves reasoning capability. The relevant artifact facts are baseline accuracy 0.0, ablation accuracy 0.0, and accuracy drop 0.0 in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/metrics.json` and `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/checker_result.json`. Because no baseline prompt was correct, the CI `[0.0, 0.0]` in `experimental/residual_migration/phase0/results/rm_phase0_20260508T161556Z/bootstrap_ci.json` is a replicate-pass diagnostic, not capability-preservation evidence.

The result is single-model, 12-trace, and Phase-0-only. It does not yet include larger frozen slices, seed repeats beyond the recorded prompt bootstrap, cross-family falsification, oracle/headroom diagnostics, latency measurements, or a deployable compression method.